

Aswin VATTAPPARAMBATHU JAYAPRAKASH

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EDUCATION

IMLEX - MSc Engineering, Lighting and XR MSc in Optics, Image, Computer Vision, Machine Learning and Multimedia MSc of Computer Science •	Sep 2024 - Sep 2026
University of Eastern Finland Universite Jean-Monnet Toyohashi University of Technology • Finland France Japan	
- Multidisciplinary program combining image processing, lighting design, and XR development across 3 institutions - Master's Thesis (Upcoming): "Industrial Training Using VR/AR-Based Prototyping and AI" at National Institute of Advanced Industrial Science and Technology (AIST), Japan (Mar–Aug 2026)	
BSc (honors) in Physics • Hindu College, University of Delhi • Delhi, India	Nov 2020 - Jul 2023

EXPERIENCE

XR Research and Development Intern	
LIRIS (Laboratoire d'InfoRmatique en Image et Systèmes d'information) • Saint-Étienne, France	Jun 2025 - Sep 2025
- Enhanced open-source 6DoF VR Thermal Feedback System (post-IEEE VR 2025 publication) by implementing obstacle-aware thermal rendering using Unity raycasting and collision detection; merged contributions to main repository, improving system realism for future XR haptics research - Integrated Skinitic Haptic Vest with custom vibrotactile feedback patterns synchronized to gameplay events (rain simulation with distributed torso pulses, vehicle close-passes with lateral vibrations, collision feedback with impact pulses) - Developed multisensory VR demo combining thermal lamps, haptic vest, and spatialized audio (via ASIO Audio Manager plugin) in a road-crossing scenario; user testing confirmed enhanced immersion compared to baseline VR-only experience - Improved developer experience by adding intuitive material system (color-coded thermal sources), teleport navigation, and hit-feedback visualization to demo scenes	

PROJECTS

Plank Experiment: VR Perception Study with Physical Props Unity, XR Interaction Toolkit, OpenXR	Nov 2025 - Nov 2025
Toyohashi University of Technology, Japan	
Link: https://aswin-avj.github.io/portfolio/projects_page/PlankExp/PlankExp.html	
- Developed Unity-based experimental framework for XR perception research investigating how physical realism influences height perception and fear responses in VR (team of 4, responsible for all VR interactions and data collection systems) - Engineered randomized trial system with elevator mechanics and height-adjustable plank scenarios. - Implemented automated data logging for subjective height judgments, movement timing, and behavioral metrics; results showed physical plank lowered Point of Subjective Equality (PSE) and increased cautious movement time	
Accessible VR Experience for Deaf/Hard-of-Hearing Users WebXR, Three.js, Gamepad API	May 2025 - Jun 2025
Jean Monnet University, France	
Link: https://aswin-avj.github.io/portfolio/projects_page/VR_sound_vis.html	
- Architected WebXR application using Three.js and JavaScript that converts non-verbal audio (birdsong, ambient sounds) into 3D directional indicators and real-time spectrograms for accessibility - Ideated and integrated multisensory feedback system combining visual audio spectrum (pitch/volume), directional proximity indicators, and haptic controller vibrations based on sound source distance - Optimized WebGL rendering pipeline for smooth performance on VR headsets; implemented JSON-based configuration system for environment customization	
AI-Based Lighting Transfer Using Synthetic Data PyTorch, Blender, Git	Mar 2025 - May 2025
Jean Monnet University, France	
Link: https://aswin-avj.github.io/portfolio/projects_page/f2b.html	
- Synthesized photorealistic dataset of 1,600 paired samples (16 face models × 100 lighting/viewpoint conditions) using 3D reconstruction tools (KeenTools FaceBuilder, TRELIS AI) and consistent rendering pipeline - Designed and trained U-Net architecture with ResNet34 encoder for facial relighting tasks; implemented hybrid loss function (MSE + SSIM) achieving 15% SSIM improvement over 500 training epochs - Validated end-to-end pipeline demonstrating successful lighting transfer (shadows, highlights, light direction) from facial images to reference sphere	

***Additional projects and professional experience available at:**

aswin-avj.github.io/portfolio

SKILLS

Programming languages & Frameworks: Python, C++, C#, Scilab (Matlab alternative), Unity (C#), WebXR, Three.js, PyTorch, Keras, ROS 2, SLAM

XR/Graphics: OpenXR, XR Interaction Toolkit, Meta Quest SDK, GLSL Shader Programming, WebGL, Real-time 3D Visualization

Tools: Git/GitHub, CUDA, LaTeX, DOTween, DaVinci Resolve

Languages: English (Bilingual), Hindi (Bilingual), Malayalam (Native), French (Beginner), Japanese (Beginner)

VOLUNTEERING & LEADERSHIP

Panchtatva, Environment Society Jan 2022 - Jan 2023

Media Head • Delhi

- Managed video production and graphic design for sustainability campaigns reaching 10,000+ audience members

Eureka, Physics Society Jan 2022 - Jan 2023

Head Member • Delhi

- Led video creation for AI and astronomy workshops; co-organized scientific literacy events